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MIL-DTL-48561B(AR)
23 March 2022
SUPERSEDING
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29 June 1988

MILITARY SPECIFICATION

DI-2-ETHYLHEXYL AZELATE

MIL-DTL-48561, dated 23 March 2022, is hereby reinstated and may be used for acquisition.

This specification is approved for use by the U.S. Army Combat Capabilities Development Command – Armaments Center (DEVCOM – AC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the manufacture and acceptance of three types of di-2-ethylhexyl azelate (see 6.1).

Type I – For general use.
Type II – For use in propellant for the M864 system.
Type III – For use in propellant for the XM1128.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM – AC, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D92	-	Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester
ASTM D97	-	Standard Test Method for Pour Point of Petroleum Products
ASTM E168	-	Standard Practices for General Techniques of Infrared Quantitative Analysis
ASTM E203	-	Standard Test Method for Water Using Volumetric Karl Fischer Titration
ASTM E260	-	Standard Practice for Packed Column Gas Chromatography
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals.
ASTM D445	-	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
ASTM D482	-	Standard Test Method for Ash from Petroleum Products
ASTM D891	-	Standard Test Method for Ash from Petroleum Products
ASTM D1193	-	Standard Specification for Reagent Water
ASTM D1209	-	Standard Test Method for Color of Clear Liquids (Platinum-Cobalt Scale)
ASTM D1218	-	Standard Test Methods for Specific Gravity, Apparent, of Liquid Industrial Chemicals
ASTM D4052	-	Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter

(Copies of these documents are available online at https://www.astm.org/.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 3961 - Animal and vegetable fats and oils – Determination of iodine value

(Copies of this documents are available online at https://www.iso.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see paragraph 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Physical and chemical requirements. The di-2-ethylhexyl azelate shall conform to the applicable physical and chemical characteristics as listed in Table I.

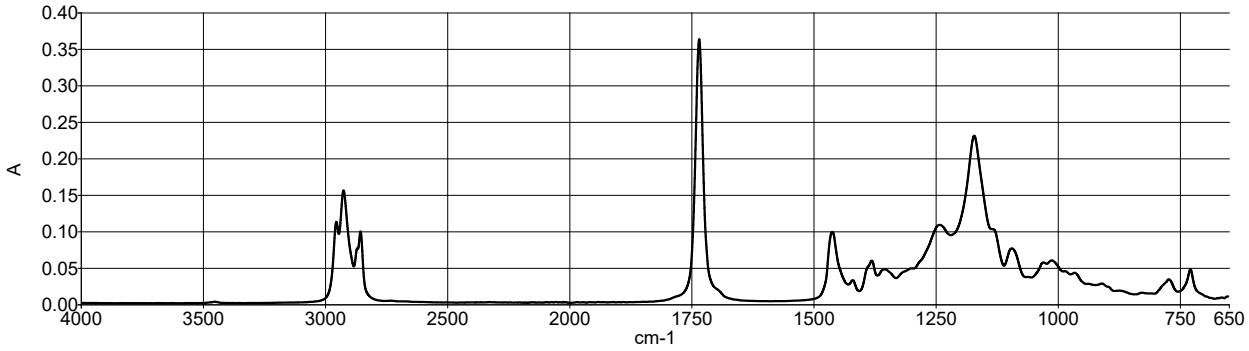
TABLE I. Physical and chemical requirements for di-2-ethylhexyl azelate.

Property	Type I	Type II	Type III
Specific gravity at 15.6˚C at 25˚C	 0.919 – 0.923 -----	 ----- 0.914 – 0.918	 ----- 0.913 – 0.919
Pour point, ˚C maximum (max.)	62.22	-----	-----
Flash point, ˚C max.	205	-----	-----
Fire point, ˚C max.	230	-----	-----
Viscosity, centistokes, at 40˚C minimum (min.) at -40˚C min.	 10.9 1250	 ----- -----	 ----- -----
Ash, % max.	0.0005	-----	-----
Iodine number max.	1.0	-----	-----
Saponification number	264 – 284	-----	-----
Boiling point, ˚C, at 5 mm Hg	-----	110 – 121	-----
Refractive index, at 25˚C	-----	1.446 – 1.448	1.447 – 1.450
Moisture, % max.	-----	-----	0.10
2-ethylhexanol content, % max.	-----	-----	0.50
Acid value, mg KOH/mg max.	0.20	-----	1.0
Note: “-----“ indicates that there is no requirement.			

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TABLE I. Physical and chemical requirements for di-2-ethylhexyl azelate. - Continued

Property	Type I	Type II	Type III
Infrared spectrum	-----	Shall contain major peaks, only, as shown in Figure 1.	
Color	No darker than standard.	White to slightly yellow	500 max., APHA
Note: "-----" indicates that there is no requirement.			



Major peaks located approximately at 2960, 2940, 2860, 1740, 1460, 1380, 1180, and 725 cm^{-1}. Some shift is expected due to instrument conditions.

FIGURE 1. Infrared (IR) spectrum for di-2-ethylhexyl azelate.

3.3 Workmanship. Di-2-ethylhexyl azelate shall be uniform in quality, free from suspended matter, and free from visible impurities and foreign material such as dirt, wooden particles or any other foreign matter.

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4. VERIFICATION

TABLE II. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration		3 – Examination 4 – Test				Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2	Specific gravity	X			X	X	X	4.4.1
3.2	Pour point	X			X	X	X	4.4.2
3.2	Flash point	X			X	X	X	4.4.3
3.2	Fire point	X			X	X	X	4.4.4
3.2	Viscosity	X			X	X	X	4.4.5
3.2	Ash	X			X	X	X	4.4.6
3.2	Iodine number	X			X	X	X	4.4.7
3.2	Saponification number	X			X	X	X	4.4.8
3.2	Boiling point	X			X	X	X	4.4.9
3.2	Refractive index	X			X	X	X	4.4.10
3.2	Moisture	X			X	X	X	4.4.11
3.2	2-ethylhexanol content	X			X	X	X	4.4.12
3.2	Acid value	X			X	X	X	4.4.13
3.2	Infrared spectrum			X		X	X	4.4.14
3.2	Color			X		X	X	4.4.15
3.3	Workmanship			X		X	X	4.5

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2).
b. Conformance inspection (see paragraph 4.3).

4.2 First article inspection. When specified (see paragraph 6.2), a sample shall be subjected to first article verification in accordance with Tables I and III. Samples of not less than two pounds of di-2-ethylhexyl azelate collected in accordance with 4.3.2 shall be submitted for examination. Samples shall be produced using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

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TABLE III. First article acceptance tests.

Category	Examination or test	Requirement paragraph	Inspection method reference
Critical	None defined		
Major			
101	Specific gravity	3.2	4.4.1
102	Pour point	3.2	4.4.2
103	Flash point	3.2	4.4.3
104	Fire point	3.2	4.4.4
105	Viscosity	3.2	4.4.5
106	Ash	3.2	4.4.6
107	Iodine number	3.2	4.4.7
108	Saponification number	3.2	4.4.8
109	Boiling point	3.2	4.4.9
110	Refractive index	3.2	4.4.10
111	Moisture	3.2	4.4.11
112	2-ethylhexanol content	3.2	4.4.12
113	Acid value	3.2	4.4.13
114	Infrared spectrum	3.2	4.4.14
115	Color	3.2	4.4.15
Minor			
201	Workmanship	3.3	4.5

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements, the first article shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph “Formation and identification of lots or batches.” Each batch shall consist of a quantity of di-2-ethylhexyl azelate that has been subjected to the same chemical and physical mixing process intended to make the final product homogeneous. A lot shall consist of one or more acceptable batches of di-2-ethylhexyl azelate produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions.

4.3.2 Di-2-ethylhexyl azelate sampling.

4.3.2.1 Sampling at the manufacturer. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300 for liquids. A composite sample of not less than 500 grams shall be formed which is assumed to be representative from a single batch. All composite samples shall be verified in accordance with 4.4. If any sample fails to comply with any applicable requirements the lot shall be rejected.

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4.3.2.2 Sampling at the received facility. Di-2-ethylhexyl azelate batches shipped in containers (e.g., drums, totes, boxes, etc.) shall be sampled and accepted using Verification Level I of MIL-STD-1916, where one unit is defined as one container. Each container shall be sampled in three (3) locations of the shipping containers in accordance with ASTM E300 for liquids. This material shall be placed in a secondary container and mixed to form a composite sample consisting of a minimum of 500 grams for each di-2-ethylhexyl azelate lot received. There shall be no reduced sampling. Every lot received shall be tested and verified in accordance with 4.3.3 and 4.4. If any sample fails to comply with any applicable requirements the lot shall be rejected.

4.3.3 Classification of characteristics. Di-2-ethylhexyl azelate shall be subjected to the inspections according to 4.3.3.1.

4.3.3.1 Di-2-ethylhexyl azelate.

Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Specific gravity	4.3.2	3.2	4.4.1
102	Pour point	4.3.2	3.2	4.4.2
103	Flash point	4.3.2	3.2	4.4.3
104	Fire point	4.3.2	3.2	4.4.4
105	Viscosity	4.3.2	3.2	4.4.5
106	Ash	4.3.2	3.2	4.4.6
107	Iodine number	4.3.2	3.2	4.4.7
108	Saponification number	4.3.2	3.2	4.4.8
109	Boiling point	4.3.2	3.2	4.4.9
110	Refractive index	4.3.2	3.2	4.4.10
111	Moisture	4.3.2	3.2	4.4.11
112	2-ethylhexanol content	4.3.2	3.2	4.4.12
113	Acid value	4.3.2	3.2	4.4.13
114	Infrared spectrum	4.3.2	3.2	4.4.14
115	Color	4.3.2	3.2	4.4.15
Minor				
201	Workmanship	4.3.2	3.3	4.5

4.4 Methods of inspection. All tests given in this section shall be performed using prescribed procedures for replicate determination given in standard analytical textbooks. Also, unless otherwise specified herein, all chemicals and reagents used in the tests shall be ACS Grade or Reagent Grade. Purified and distilled water in accordance with ASTM D1193 shall be used as required in all test procedures.

4.4.1 Specific gravity. Determine specific gravity of the di-2-ethylhexyl azelate in accordance with the following methods, as applicable.

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4.4.1.1 Specific gravity, at 15.6˚C (Type I). Specific gravity of Type I material shall be determined in accordance with ASTM D891.

4.4.1.2 Specific gravity, at 25˚C (Type III). Determine specific gravity of the di-2-ethylhexyl azelate at 25˚C in accordance with ASTM D4052.

4.4.2 Pour point, ˚C. Determine pour point in accordance with ASTM D97.

4.4.3 Flash point, ˚C. Determine flash point in accordance with ASTM D92.

4.4.4 Fire point, ˚C. Determine fire point in accordance with ASTM D92.

4.4.5 Viscosity. Determine viscosity in accordance with ASTM D445.

4.4.6 Ash. Determine ash content in accordance with ASTM D482.

4.4.7 Iodine number. Determine iodine number in accordance with ISO 3961.

4.4.8 Saponification number. Transfer a 2.0 g sample, weighed to 0.2 mg, to a 250-mL Erlenmeyer flask (24/40 ground glass joint) and 50.00 mL of approximately 0.5% alcoholic potassium hydroxide solution. Carry along a blank determination on 50.00 mL of alcoholic potassium hydroxide solution. Reflux under a water-cooled condenser on an electric hot plate for 2 hours. Allow the flask and contents to cool to room temperature. Add 30 mL of 95% ethyl alcohol through the top of the condenser and disconnect the condenser. Add 3 drops of 1% phenolphthalein solution and titrate with standard 0.5N hydrochloric acid to the disappearance of the pink color. Calculate the saponification number as follows:

$$\text{Saponification number} = \frac{56.1 \times (B - A) \times N}{W}$$

Where: A = mL of hydrochloric acid required for sample
B = mL of hydrochloric acid required for blank
N = normality of hydrochloric acid
W = grams of sample

4.4.9 Boiling point, at 5 mm Hg.

4.4.9.1 Apparatus.

a. 500-mL, 3 neck flask
b. Vacuum distilling apparatus
c. Glas-Col heating unit
d. Variable transformer
e. Ground glass thermometer (range 10-360˚C, subdivisions of 1˚C)
f. Glass wool insulating material
g. Sources of vacuum, 110 AC electricity, cold water (60-70˚C)
h. Bleed-off apparatus to control pressure (stop clock)

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i. Manometer accurate to 0.1 mm Hg
j. Flexaframe and support clamps

4.4.9.2 Procedure.

a. Weigh out 500 g of sample and record to nearest 0.001 g.
b. Assemble distillation apparatus on the Flexaframe. Assemble Glas-Col heater independently so it can be removed from flask. Insulate flask and distillation apparatus with glass wool.
c. Pour sample into flask and insert thermometer in a position to measure vapor temperature.
d. Apply vacuum and regulate to desired pressure (5mm on di-2-ethylhexyl azelate) using bleed-off and measure with manometer.
e. After the temperature of vapor has stabilized, record temperature to nearest 1˚C and the pressure at which it was obtained. Report for conformance.

4.4.10 Refractive index. Determine refractive index in accordance with the following methods, as applicable.

4.4.10.1 Refractive index, Type I.

4.4.10.1.1 Apparatus.

a. Abbe-3L refractometer or equivalent
b. Solvent for sample
c. Soft tissues
d. Acetone

4.4.10.1.2 Procedure. (See 6.6).

a. Clean the upper and lower prisms with soft, non-abrasive, absorbent tissues wetted with acetone; allow them to dry.
b. Introduce the sample upon the prism by means of a pipette or dropper. Only a few drops are required. The prisms are then closed slowly allowing the excess to squeeze out into the space between the metal mounts.
c. After the sample is in position on the instrument, set the scale at the approximate value expected.
d. Release the switch. Bring the borderline which will probably be strongly colored, near the crosshair and compensate the color by adjusting the position of the dial (see Figure 1).
e. Observe the crosshair, sharply focusing the eyepiece if necessary and bring the dividing line upon their intersection by means of the course of fine hand controls.
f. Read the refractive index by depressing the momentary contact switch and report for conformance.

4.4.10.2 Refractive index, Type III. Verify refractive index in accordance with ASTM D1218.

4.4.11 Moisture. Verify moisture content in accordance with ASTM E203.

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4.4.12 2-ethylhexanol content. Verify 2-ethylhexanol content in accordance with ASTM E260.

4.4.13 Acid value. Transfer approximately 100 mL of a mixture of equal parts by volume of 95% alcohol and reagent grade benzene to a 300-mL Erlenmeyer flask. Heat the flask and contents to incipient boiling. Add 3 drops of 1% phenolphthalein solution and titrate with standard 0.01N sodium hydroxide solution to a faint pink color. Add 50-mL of the sample and titrate again to a faint pink color. Calculate the acid number as follows:

$$\text{Acid number} = \frac{56.1 \times A \times N}{M \times D}$$

Where: A = mL of sodium hydroxide solution required for sample
N = normality of sodium hydroxide solution
M = mL of sample taken
D = specific gravity of sample

4.4.14 Infrared spectrum. Verify by examination that an infrared spectrum collected in accordance with ASTM E168 matches a reference standard of di-2-ethylhexyl azelate.

4.4.15 Color. Determine color in accordance with the following methods as appropriate for Type.

4.4.15.1 Color (Type I). Add 1.0 mL of 0.01N potassium dichromate (prepared by dissolving 0.4904 g of $K_2Cr_2O_7$ in water and diluting to 1 L) to 49.0 mL of water in a Nessler tube and compare the color with 50 mL of specimen in a Nessler tube.

4.4.15.2 Color (Type II). Determine by visual examination that the sample complies with the requirement.

4.4.15.3 Color (Type III). Determine color in accordance with ASTM D1209.

4.5 Workmanship. Place approximately 100 mL of di-2-ethylhexyl azelate into a 150 mL glass beaker over a clean, well-lighted white surface. The sample shall be examined by personnel with normal vision. The sample shall be uniform in quality and free from visible impurities, and foreign material such as dirt, wooden particles or any other foreign matter. Failure to meet the workmanship requirement during inspection or while executing production processes shall result in rejection of the batch/lot represented (see 4.3.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the

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Inventory Control Point' packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The di-2-ethylhexyl azelate (commonly called dioctyl azelate) covered in this specification is intended for use in lubricants (Type I); for use in M864 propellant (Type II); and for use in XM1128 propellant (Type III).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Whether a first article sample is required (see 3.2).
c. Packaging requirements (see 5.1).
d. Whether acceptance and description sheets are required (see 6.2.1)

6.2.1 Data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, acceptance and description sheets prepared in accordance with MIL-STD-1171, or as directed in the contract should be supplied by the contractor.

6.3 Submission of inspection equipment designs for approval. Submit equipment designs, as required, to the Technical Agency (see 6.4)

6.4 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM – AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.5 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to the Technical Agency (see 6.4). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.6 Special notes on refractometers.

a. The refractive index decreases with a temperature rise because the density decreases. The prisms can be temperature controlled to meet the need.
b. Free-flowing liquids can be introduced with an eyedropper into the channel which is alongside the prisms. When a mixture to be analyzed contains a volatile component, this technique is particularly useful because it minimizes the loss by

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evaporation of the volatile component and thus affects the accuracy of the readout.

c. Never touch the prisms with any hard object. Always clean with soft tissues moistened with alcohol, acetone, or ether. After cleaning, always leave the prisms in a locked position.

6.7 Subject term (key word) listing.

Plasticizer
Igniter
Base bleed
Propellant
Burn rate modifier
M864
XM1128
Dioctyl azelate

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
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Preparing activity:
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